

**Classified**



## **Geotopia Police Department Violation Record**

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# Geotopia Police Department Violation Record



# 2

**Name:**

John Tripleang

**Address:**

154 Tangram St.  
Geotopia, GT 11223

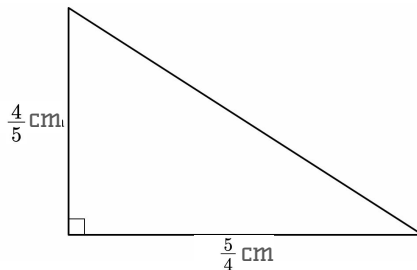
**ID Number:**

3062471572

**Photo:**



"What is the area, in Square centimeters, of the shape below?  
Express your answer as a fraction in simplest form."



**Details of Violation:**

~~John answered:~~

~~Length:  $\frac{5}{4}$  cm~~

~~Width:  $\frac{4}{5}$  cm~~

~~Area = Length times width~~

~~$$\frac{5}{4} \cdot \frac{4}{5} = \frac{20}{20} = 1 \text{ cm}$$~~

**Officer's note:**

1) We call the sides of a triangle "base" and "height", not "length and width".

2) Area of a triangle is  $\frac{1}{2}bh$ . So it should be:  $\frac{1}{2} \cdot \frac{5}{4} \cdot \frac{4}{5} = \boxed{\frac{1}{2}}$

3) All area units are with "square". So the answer should be  $\boxed{\frac{1}{2} \text{ cm}^2}$



# Geotopia Police Department Violation Record



# 3

**Name:**  
Victoria Tomoro

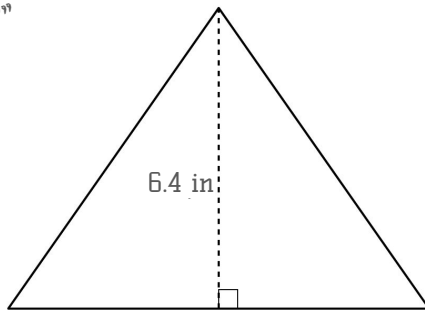
**Address:**  
39 Triangle Ave.  
Geotopia, GT 10066

**ID Number:**  
6775503923

**Photo:**



"The area of the triangle below is 28.8 Square inches. What is the length of the base?"



**Details of Violation:**

~~Victoria's answer:~~

$$\frac{28.8}{6.4} = 4.5 \text{ in}^2$$

**Officer's note:**

1) Use area formula to solve:

$$A = \frac{1}{2} \cdot b \cdot h$$

$$28.8 = \frac{1}{2} \cdot 6.4 \cdot h$$

$$28.8 = 3.2 \cdot h$$

$$9 = h$$

2) Base or height is a "line", not "area" so the unit should be 9 in



# Geotopia Police Department Violation Record



4

Name:

Ada Saito

Address:

2699 Trapezoid Ave.  
Geotopia, GT 01183

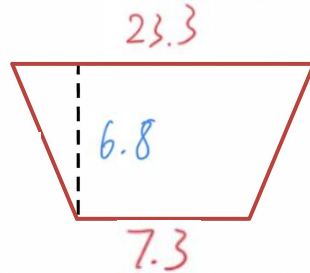
ID Number:

1773811290

Photo:



"What is the total area, in Square inches, of the shaded sections of the trapezoid below?"



Details of Violation:

~~Ada's answer:~~

~~Base: 23.3~~

~~Height: 6.8~~

~~Area:  $\frac{1}{2}bh$~~

~~$\frac{1}{2} \cdot (23.3) \cdot (6.8) = 79.22 \text{ in}^2$~~

Officer's note:

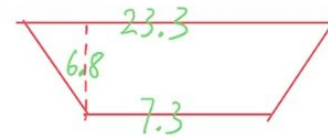
Area of a Trapezoid is

$$A = \frac{1}{2}(b_1 + b_2)h$$

There are 2 bases in a trapezoid. They are always parallel to each other.

base 1: 23.3

base 2: 7.3



$$\text{Area} = \frac{1}{2}(b_1 + b_2)h$$

$$= \frac{1}{2} \cdot (23.3 + 7.3) \cdot 6.8 = 104.04$$

The area is 104.04 in<sup>2</sup>



# Geotopia Police Department Violation Record



5

Name:

Thomas Vossenbergl

Address:

86 Tangram St.  
Geotopia, GT 11223

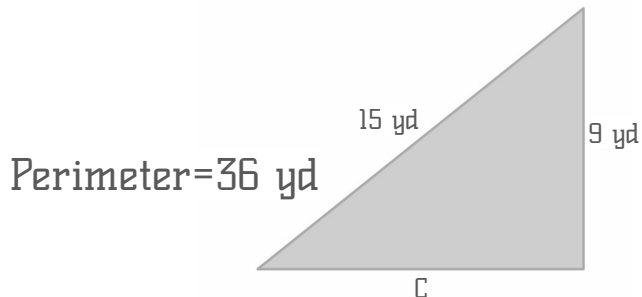
ID Number:

1381077562

Photo:



"Given the perimeter of this triangle which is 36 yards, find the value of c."



Details of Violation:

~~Thomas's answer:~~

~~$$\frac{36}{9} = 4 \text{ yd}$$~~

Officer's note:

This is perimeter! Not area!

Add all 3 sides up for the perimeter.

$$15 + 9 + c = 36$$

$$\begin{array}{r} 24 + c = 36 \\ -24 \quad -24 \\ \hline \end{array}$$

$$c = 12$$

c is 12 yards long.



# Geotopia Police Department Violation Record



6

Name:  
Tyrone Atkins

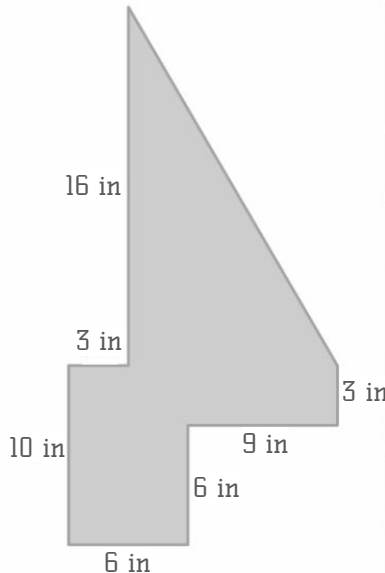
Address:  
525 North GeoPlaza  
Geotopia, GT 12355

ID Number:  
0989952108

Photo:



"what is the  
area of this  
polygon?"

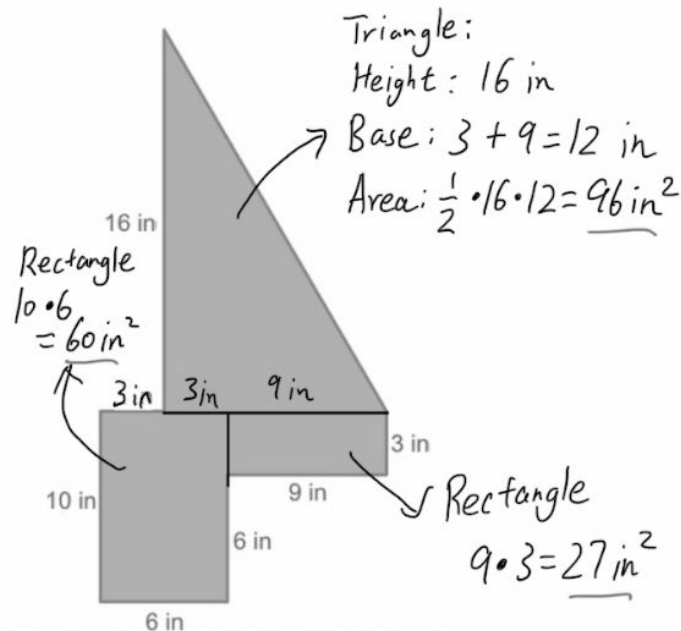


Details of Violation:

~~Tyrone's answer:~~

$$\begin{aligned} &16 + 2 + 3 + 9 + 6 + 6 + 10 \\ &= 52 \text{ in}^2 \end{aligned}$$

Officer's note:



Triangle:  
Height: 16 in  
Base:  $3 + 9 = 12 \text{ in}$   
Area:  $\frac{1}{2} \cdot 16 \cdot 12 = \underline{96 \text{ in}^2}$

Rectangle  
 $10 \cdot 6$   
 $= \underline{60 \text{ in}^2}$

Rectangle  
 $9 \cdot 3 = \underline{27 \text{ in}^2}$

$$\begin{aligned} \text{Total Area: } &96 + 60 + 27 \\ &= \underline{183 \text{ in}^2} \end{aligned}$$



# Geotopia Police Department Violation Record



# 7

**Name:**  
Harry Lopez

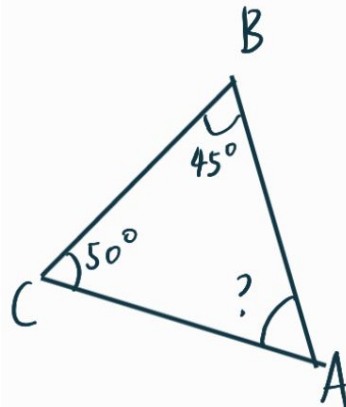
**Address:**  
530 South GeoPlaza  
Geotopia, GT 12355

**ID Number:**  
1989954448

**Photo:**



"How do you find the measure of angle A?"



**Details of Violation:**

~~$$50 + 45 = 95$$~~

~~$$50 - 45 = 5$$~~

**Officer's note:**

Remember, all 3  
angles inside  
a triangle always  
add up to **180°**

$$\begin{aligned}\angle A &= 180^\circ - \angle B - \angle C \\ (\text{Angle A}) &= 180^\circ - 45^\circ - 50^\circ \\ &= \underline{85^\circ}\end{aligned}$$



# Geotopia Police Department Violation Record



8

Name:  
Jenni Angelo

Address:  
1177 GeoTown st  
Geotopia, GT 09985

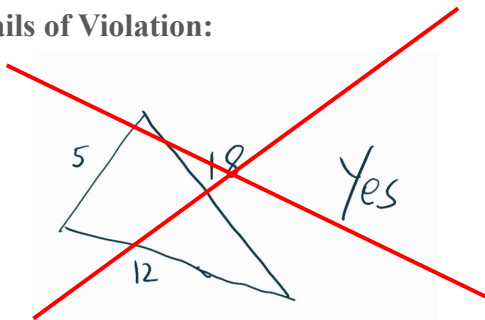
ID Number:  
4340433448

Photo:



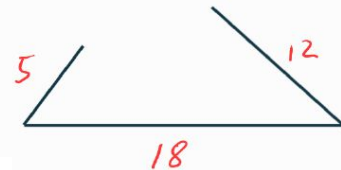
"Can 5, 12 and 18  
make a  
triangle?"

Details of Violation:



Officer's note:

No, 5, 12 and 18 cannot  
be the side lengths of  
a triangle!



It is because  $5+12=17$  which  
is less than 18. The sides  
of 5 and 12 are not long  
enough to reach each other!  
Also,  $18-12$  is 6 which  
is greater than 5. We need  
the difference to be less than  
the third side.



# Geotopia Police Department Violation Record



9

Name:  
Haru Palumbo

Address:  
1177 GeoTown st  
Geotopia, GT 09985

ID Number:  
4340433448

Photo:



"How many possible lengths of the third side if I have two sides of 5 and 12 to make a triangle?"

Details of Violation:

~~A lot!  $12 - 5 = 7$   
So 7 possible lengths!~~

Officer's note:

$$5 + 12 = 17 \quad (\text{Sum of two sides})$$

$$12 - 5 = 7 \quad (\text{Difference of two sides})$$

So, any number between 7 and 17 will work.  
(Exclude 7 and 17)

8, 9, 10, 11, 12,  
13, 14, 15, 16  
Any of these are possible!



# Geotopia Police Department Violation Record



# 10

Name:

Nelson Wu

Address:

3217 Tangram St.  
Geotopia, GT 11223

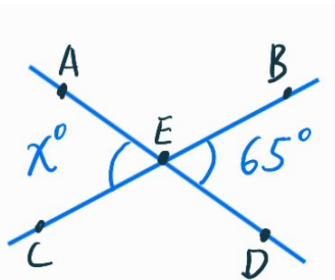
ID Number:

4340111338

Photo:



"What is the  
value of  $x$ ?"



Officer's note:

$\angle AEC$  and  $\angle BED$   
are **vertical angles**  
because they are  
opposite to each other and  
share the same vertex.  
**Vertical angles are**  
**EQUAL** in measure.  
 $x = 65$

Details of Violation:

~~$x + 65 = 65x$~~



# Geotopia Police Department Violation Record



# 11

Name:  
Veronika Siabs

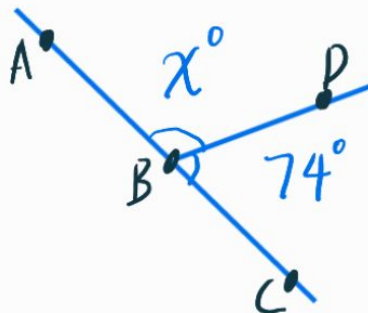
Address:  
13 Triangle Ave.  
Geotopia, GT 10066

ID Number:  
1233633448

Photo:



"What is the value of  $x$ ?"



Officer's note:

$\angle ABD$  and  $\angle DBC$   
are *supplementary*  
*angles* because they  
add up to  $180^\circ$   
(Measure of a "flat"  
angle)

To solve for  $x$ :

$$x + 74 = 180$$

$$x = 106$$

Details of Violation:

~~$x = 74^\circ$~~



# Geotopia Police Department Violation Record



# 12

Name:  
Gigi Madkins

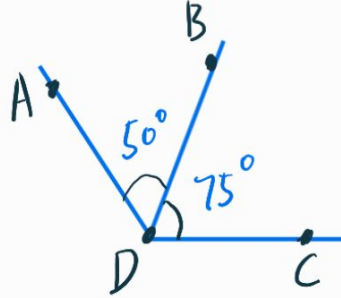
Address:  
512 Vertikal Avenue  
Geotopia, GT 15440

ID Number:  
2768122448

Photo:



"What is the  
angle  
relationship?"



Details of Violation:

~~$\angle ADB$  and  $\angle BDC$   
are supplementary  
angles.~~

Officer's note:

$\angle ADB$  and  $\angle BDC$   
are NOT supplementary  
because they do not  
add up to  $180^\circ$ .  
They are adjacent  
angles because they  
share a side.



# Geotopia Police Department Violation Record



# 13

Name:

Amani Verez

Address:

18 Supple Avenue  
Geotopia, GT 15430

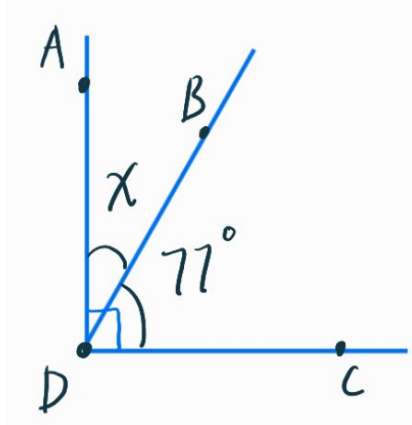
ID Number:

2768129878

Photo:



"What is the value of  $x$ ?"



Details of Violation:

~~$$x + 77 = 180$$~~

~~$$x = 103$$~~

Officer's note:

$\angle ABD$  and  $\angle BDC$   
are **Complementary angles**  
because they add up  
to  $90^\circ$  (right angle)  
To solve for  $x$ :

$$x + 77 = 90$$

$$x = 90 - 77 = 23$$



# Geotopia Police Department Violation Record



# 14

**Name:**  
Eugene Johnson

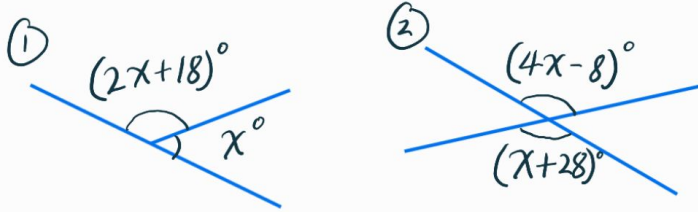
**Address:**  
99 Adjacent street  
Geotopia, GT 22440

**ID Number:**  
2768333448

**Photo:**



"What is the value of x?"



**Details of Violation:**

~~①  $2x+18=x$~~   
~~②  $4x-8+x+28=180$~~

**Officer's note:**

① Supplementary  
 $2x+18+x=180$   
 $3x+18=180$   
 $3x=162$   
 $x=54$

② Vertical  
 $4x-8=x+28$   
 $4x-x=8+28$   
 $3x=36$   
 $x=12$



# Geotopia Police Department Violation Record



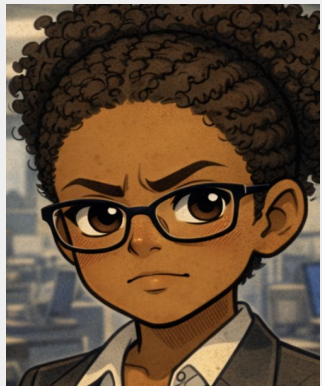
15

Name:  
Alina Mbappe

Address:  
112 Circle way  
Geotopia, GT 22410

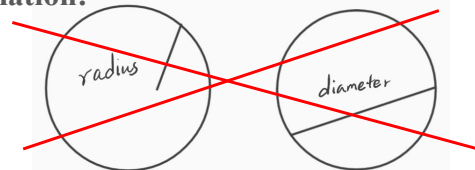
ID Number:  
1751333448

Photo:



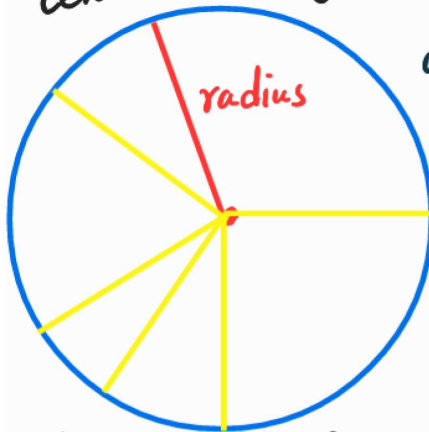
"Draw a radius and a diameter"

Details of Violation:



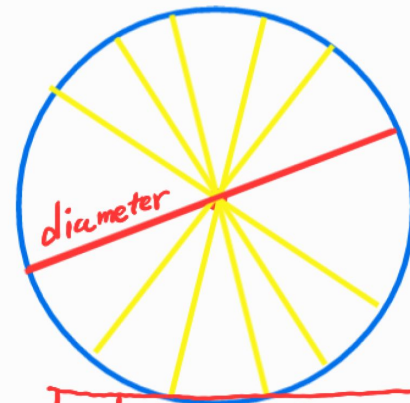
Officer's note:

Radius is from  
center to edge



You can draw infinite  
amount of radius and diameter

Diameter is from  
edge to center to edge



$$d = 2 \cdot r$$

diameter is twice radius



# Geotopia Police Department Violation Record



# 16

Name:  
Omar Hassan

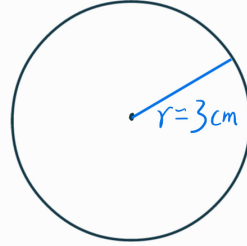
Address:  
32 Area Road  
Geotopia, GT 12410

ID Number:  
1743883448

Photo:



"What is the  
circumference if  
the radius is 3cm?"



Details of Violation:

~~$$3 \cdot \pi = 3\pi \text{ cm}$$
$$3 \cdot 3.14 = 9.42 \text{ cm}$$~~

Officer's note:

If you know the radius,  
use  $C = 2\pi r$

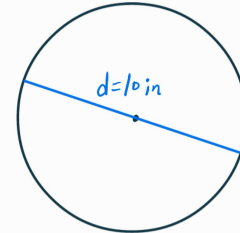
Circumference is double radius  
times  $\pi$

$$C = 2 \cdot \pi \cdot 3$$
$$= 6\pi = 6 \cdot 3.14 = 18.84$$

$6\pi \text{ cm}$  is in terms of  $\pi$   
 $18.84 \text{ cm}$  is in decimal

If you know the diameter,  
use  $C = \pi d$

Circumference is  $\pi$  times diameter.  
 $C = \pi \cdot 10 = 10\pi \text{ in} = 10 \cdot 3.14 = 31.4 \text{ in}$   
(in terms of  $\pi$ ) (in decimal)





# Geotopia Police Department Violation Record



# 17

Name:  
Hongqing Wei

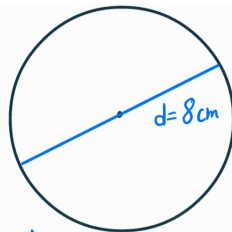
Address:  
119 Tianan Avenue  
Geotopia, GT 11310

ID Number:  
0743903440

Photo:



"What is the area if  
the diameter is  
8cm?"



Details of Violation:

~~$$\begin{aligned} \text{Area} &= 8^2 \cdot \pi \\ &= 64\pi \text{ cm} \end{aligned}$$~~

Officer's note:

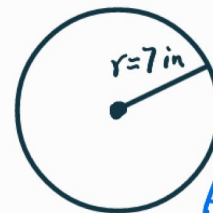
We always need radius  
to solve for area!

$$A = \pi r^2$$

Area is  $\pi$  times radius square,  
 $d = 8\text{cm} \rightarrow r = 4\text{cm}$  (half)

$$\begin{aligned} A &= \pi \cdot 4^2 = 16\pi \text{ cm}^2 \\ &= 16 \cdot 3.14 \\ &= 50.24 \text{ cm}^2 \end{aligned}$$

( $16\pi$  is in terms of  $\pi$ , 50.24 is in decimal.)



We have the  
radius 7 in.

$$\begin{aligned} A &= \pi \cdot 7^2 = 49\pi \\ &= 49 \cdot 3.14 \\ &= 153.86 \end{aligned}$$

Either  $49\pi \text{ in}^2$   
or  $153.86 \text{ in}^2$



# Geotopia Police Department Violation Record



# 18

**Name:**  
Yuki Nakamura

**Address:**  
119 Tianan Avenue  
Geotopia, GT 11310

**ID Number:**  
0743903441

**Photo:**



"The circumference of a circle is 28.26 cm. What is the area?"

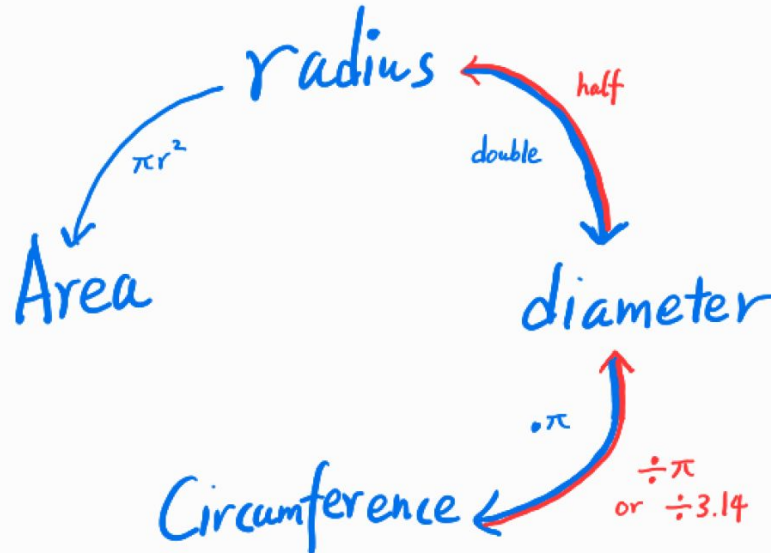
**Details of Violation:**

~~$$C = 28.26 \text{ cm}^2$$~~

~~$$d = 28.26 \div \pi = 9 \text{ cm}$$~~

~~$$A = 9^2 \cdot \pi = 254.34 \text{ cm}^2$$~~

**Officer's note:**



Follow this chart.

If we have  $C = 28.26$ ,  
then  $\div 3.14$  is 9 for diameter.

$9 \div 2 = 4.5$  for radius.

$$4.5^2 \cdot \pi = \underline{\underline{63.59 \text{ cm}^2}}$$



# Geotopia Police Department Violation Record



# 19

**Name:**  
Alejandro García

**Address:**  
89 Volume Avenue  
Geotopia, GT 12341

**ID Number:**  
0678833422

**Photo:**



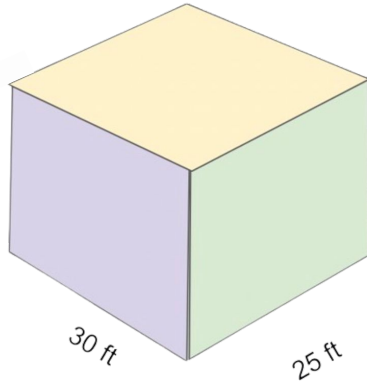
"What is the volume of the rectangular prism with 30 ft, 25 ft, and 15 ft as the length, width and height?"

**Details of Violation:**

~~Volume?~~

~~$30 + 25 + 15 = 70 \text{ ft}^3$~~

**Officer's note:**



**Volume:**

Length \* Width \* Height

**30**

\*

**25**

\*

**15**

= **11250** ft<sup>3</sup>

Volume of a prism is always the base area times the height.





# Geotopia Police Department Violation Record



# 20

**Name:**  
Aahana Gupta

**Address:**  
47 Surface Avenue  
Geotopia, GT 10011

**ID Number:**  
1238567723

**Photo:**



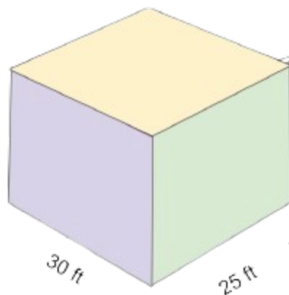
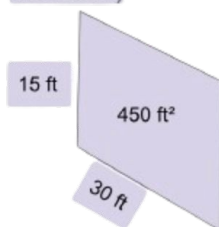
"What is the surface area of this rectangular prism which has length, width and height in 30 ft, 25 ft and 15 ft?"

**Details of Violation:**

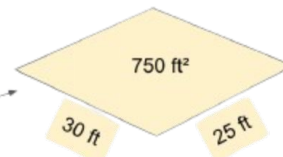
~~Surface Area?  
 $30 \times 25 \times 15 = 11250 \text{ ft}^2$~~

**Officer's note:**

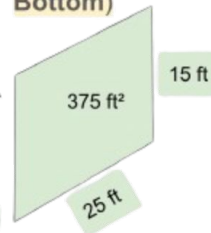
This face is the Front  
(The area is the same as the Back)



This face is the Top  
(The area is the same as the Bottom)



This face is the right  
(The area is the same as the left)



**Surface Area:**

|        |                |
|--------|----------------|
| Top    | $750 \times 2$ |
| Bottom | +              |
| Left   | $375 \times 2$ |
| Right  | +              |
| Front  | $450 \times 2$ |
| Back   |                |

|            |      |                 |
|------------|------|-----------------|
| <b>Sum</b> | 3150 | ft <sup>2</sup> |
|------------|------|-----------------|



# Geotopia Police Department Violation Record



# 21

**Name:**  
Stephenie Anderson

**Address:**  
135 Prism Avenue  
Geotopia, GT 10323

**ID Number:**  
1456119973

**Photo:**

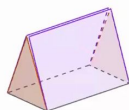
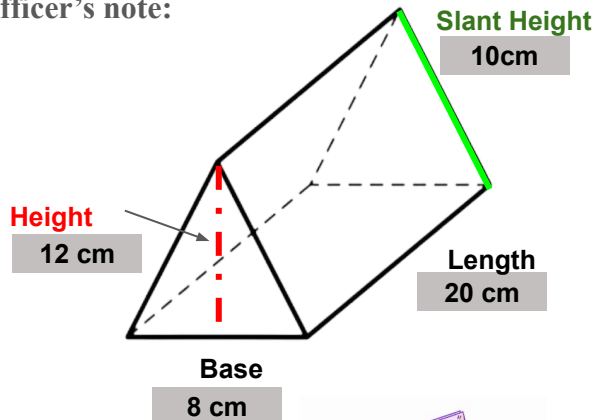


"What is the **Surface area** and **volume** of a **triangular prism** which has base of 8 cm, height of 12 cm, length of 20 cm and slant height of 10 cm?"

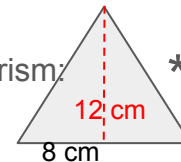
**Details of Violation:**

~~Volume:  $12 \times 8 \times 20 \times 10 = 19200 \text{ ft}^3$~~   
~~Surface Area:  $12 \times 8 \times 20 = 1920 \text{ ft}^2$~~

**Officer's note:**



**Volume of  
Triangular Prism:**



\*

Length = 20 cm

$$(\frac{1}{2} \times 8 \times 12) \times 20 = 960 \text{ cm}^3$$

**Surface Area:**

Front + Back + Right + Left + Bottom

Diagram showing the calculation of surface area by summing the areas of the front, back, right, left, and bottom faces of the prism.

$$\frac{1}{2} \times 8 \times 12 + \frac{1}{2} \times 8 \times 12 + 20 \times 10 + 20 \times 10 + 8 \times 20$$

**656** cm<sup>2</sup>



# Geotopia Police Department Violation Record



# 22

Name:  
Acozy Guy

Address:  
3089 Pyramid Street  
Geotopia, GT 28009

ID Number:  
1347890445

Photo:

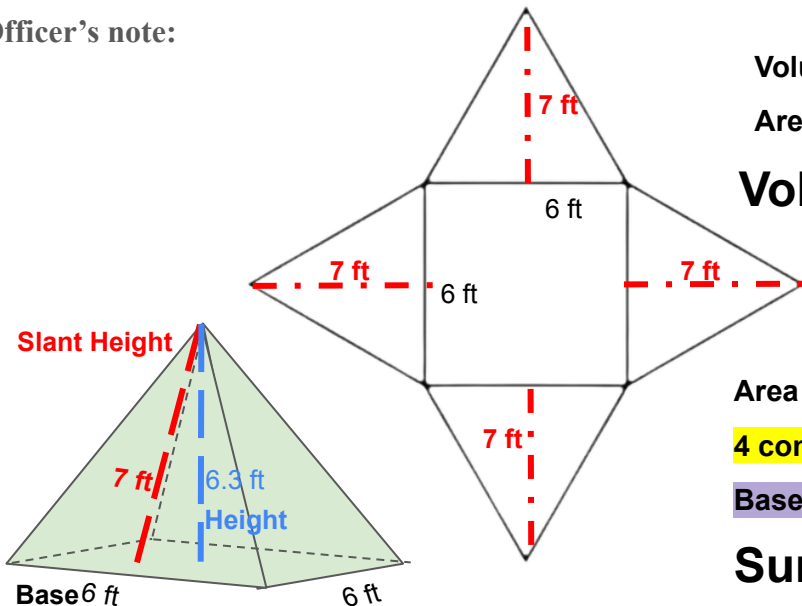


"What is the **Surface area** and **volume** of a **Square pyramid** which has base of 6 ft, height of 6.3 ft, slant height of 7 ft?"

Details of Violation:

~~Volume:  $7 \times 6 \times 6 = 252 \text{ ft}^3$   
Surface Area:  $21 \times 4 = 84 \text{ ft}^2$~~

Officer's note:



Volume is the  $\frac{1}{3}$  the base triangle times the height

Area of front triangle =  $\frac{1}{2} * 7 * 6 = 21 \text{ ft}^2$

Volume =  $\frac{1}{3} * 21 * 6.3 = 44.1 \text{ ft}^3$

Area of front triangle =  $\frac{1}{2} * 7 * 6 = 21 \text{ ft}^2$

**4 congruent triangles:**  $21 \text{ ft}^2 * 4 = 84 \text{ ft}^2$

**Base** =  $6 * 6 = 36 \text{ ft}^2$  (this is a square)

**Surface Area** = **36** + **84** = **120 ft<sup>2</sup>**



# Geotopia Police Department Violation Record



# 23

**Name:**  
Crystal Liam

**Address:**  
88 Cross Pl  
Geotopia, GT 10086

**ID Number:**  
5163036819

**Photo:**

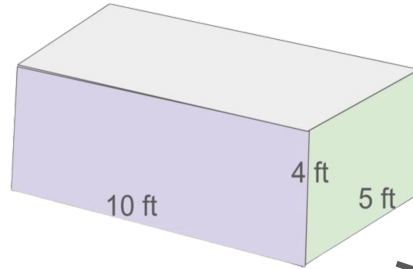


"What is the area of the  
**cross sections** ?"

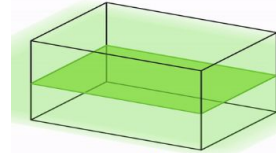
**Details of Violation:**

~~Cross Section is  
crossing a section,  
right?~~

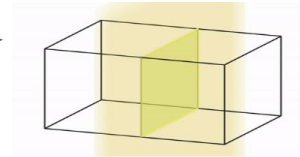
**Officer's note:**



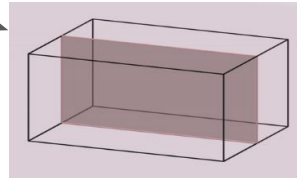
The cross section of this prism  
is a rectangle.



If we cut horizontally:  
 $10 * 5 = 50 \text{ ft}^2$



If we cut vertically:  
 $4 * 5 = 20 \text{ ft}^2$



If we cut vertically from the right:  
 $10 * 4 = 40 \text{ ft}^2$